Hamza Sadaqat

GIS & Remote Sensing Specialist · 4+ years of professional Remote Sensing, EO & GIS experience

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PROFILE

GIS and remote sensing specialist with more than four years of experience delivering geospatial solutions for agriculture, water resources, natural hazards, infrastructure, real estate and environmental monitoring. Experienced in multispectral and hyperspectral image analysis, stereo DEM production, flood and hazard mapping, machine-learning classification, spatial data engineering and automation, and in presenting results through publication-quality maps, web maps and GIS dashboards. Holds an MS in Remote Sensing & GIS (CGPA 3.96/4) and has delivered 376+ freelance projects.

EDUCATION

MS Remote Sensing & GIS 2024 – 2026
PMAS Arid Agriculture University, Rawalpindi · CGPA 3.96 / 4 · thesis on glacier area and volume change in the Hunza River Watershed

BS Geo-informatics · Silver Medalist 2018 – 2022
PMAS Arid Agriculture University, Rawalpindi · CGPA 3.81 / 4 · ranked 2nd of 45

HONOURS AND AWARDS

- **Silver Medal** (BS, 2nd of 45) · **Vice-Chancellor's Talent Scholarship** multiple times for topping semester examinations in BS and MS · **Prime Minister's Laptop Award** for academic merit.

PROFESSIONAL EXPERIENCE

GIS & Remote Sensing Specialist Oct 2025 – present
Geomapping Solutions (Pvt) Ltd, Islamabad, Pakistan

- Multispectral and hyperspectral satellite image processing for agricultural, environmental and land-use monitoring.
- Pre-processing including radiometric and atmospheric correction, georeferencing and large-area mosaicking.
- Land-cover mapping with supervised and unsupervised classification; multi-temporal change detection.
- 1 m DSM/DTM, contours and catchments from 50 cm satellite stereo pairs for hill-torrent flood modelling and dam siting; flood hazard and exposure atlas covering 2,281 km².
- Geodatabase design and management, data accuracy assessment and validation.

GIS Manager Jul 2022 – Oct 2025
Changaiz Enterprises, Rawalpindi, Pakistan

- Defined GIS requirements with end users and resolved technical issues.
- Planned and processed drone imagery and DGPS surveys for cadastral mapping, DEM/DTM and topographic maps.
- Administered the GIS server and software with IT; handled spatial and non-spatial data for large project areas.
- Validated GIS data to company SOPs and coordinated field services, satellite and aerial imagery with external stakeholders.

GIS Analyst (volunteer) Sep 2024 – Mar 2025
Tomkulak, Switzerland (remote)

- GIS support for a non-profit network working on critical-infrastructure research and solutions.

Freelance GIS Consultant 2018 – 2022
Independent (remote) · clients sourced through LinkedIn

- Delivered 376+ projects for international clients alongside my BS studies.
- Work included web apps for crop-area measurement, dashboards for a Johns Hopkins University project, drone processing in WebODM, 3D city models in CityEngine, geographically weighted regression and a Python heat-stress (WBGT) hotspot tool for ArcGIS.

Freelance Project Manager Jan 2020 – Oct 2021
Translation7, London (remote)

- Managed 50+ legal and general translation and transcription projects with linguists in 20+ languages.

Medical Transcriptionist Jan 2017 – Nov 2020
Eye Technologies (Pvt) Ltd, Rawalpindi

SELECTED PROJECTS

- **Glacier change, Hunza River Watershed (2000–2025):** glacier inventory and mass balance (−0.34 ± 0.15 m w.e. a^{-1}) from multi-sensor imagery and self-built ASTER stereo DEMs.
- **Hill-torrent flood hazard (2025–2026):** 25/50/100-year inundation, exposure of 3.31 million people and 145 health facilities, support for proposed dam sites.
- **National crop survey (2022–2023):** 136,822 field observations in 83 districts mapped by crop and season.
- **Brick kilns, Punjab (2026):** 11,272 kilns mapped; district emissions and exposure of schools and hospitals.
- **Groundwater contamination, Libya (2025):** Earth Engine pipeline with ten indices and 70 wells.
- **Hyperspectral classification (2026):** EnMAP and ROSIS land-cover mapping at 89–93% overall accuracy.

TECHNICAL SKILLS

Remote sensing	Multispectral, hyperspectral and SAR processing; radiometric & atmospheric correction; spectral indices (NDVI, NDSI, MNDWI, LST, VHI); change detection and time series; accuracy assessment
GeoAI & machine learning	Random Forest, SVM and CART classification (scikit-learn, Google Earth Engine); object-based image analysis (eCognition); training-sample design
Photogrammetry & terrain	Satellite and UAV stereo photogrammetry; DSM/DTM extraction, co-registration and DEM differencing; LiDAR point clouds; contours and catchments
Hydrology & hazards	Watershed delineation; flood inundation and exposure analysis; HEC-RAS, WEAP, MODFLOW (working knowledge); glacier mass balance; drought indices
GIS software	ArcGIS Pro, ArcGIS Online, QGIS, ERDAS IMAGINE, ENVI, SNAP, PCI Geomatica, Global Mapper, GRASS GIS, ArcGIS CityEngine, Agisoft Metashape, Pix4D, WebODM, Ames Stereo Pipeline
Programming & automation	Python (ArcPy, GDAL/OGR, rasterio, pandas, NumPy, scikit-learn), Google Earth Engine (JavaScript & Python), R, ModelBuilder, Jupyter
Web GIS & visualisation	ArcGIS Dashboards, Experience Builder, StoryMaps, Earth Engine Apps, HTML/CSS/JavaScript (three.js); publication-quality cartography
Field data	Drone (UAV) mapping, DGPS/GNSS surveys, ground control points, cadastral and topographic surveys

CERTIFICATIONS AND TRAINING

- GIS Surveyor, 360 hours, Grade A+ (PSDF / PBTE Lahore, 2019) · Microsoft Technology Associate: Introduction to Programming Using Python.
- Esri MOOCs (Spatial Analysis; GIS for Climate Action; GIS and BIM; Python) · Coursera: Geospatial and Environmental Analysis (UC Davis), Geodesign (Penn State).
- Climate Change AI Summer School (2024) · Ban Ki-moon Centre Green Jobs Mentorship, Vienna (2024) · SIRGAS Virtual School (2025) · China–Pakistan Agricultural Technology Training (2025).
- Organiser, AGRISAT international conference and training workshop on high-resolution satellite data for agriculture (2026).

PUBLICATIONS

1. **Sadaqat H.**, Muzaffar A., Muzaffar A., Sajid A., Chalchisa A., Abid K., Farooq H. (2024). *Remote sensing assessment of climate variability impact on cotton crop in Faisalabad, Sahiwal, and Pakpattan.* Journal of Ecology and Conservation, 2(3), 21–26. doi.org/10.61577/jec.2024.1000014
2. Chalchisa A., **Sadaqat H.**, Kebede T., Dessaleng A., Wuyep L.C., Sajid A. (2024). *Developing optimal road alignment using geospatial technologies: a case study from Ginchi to Chullute, Oromia region.* Journal of Geosciences Insights, 2(4), 7–13. doi.org/10.61577/jgi.2024.1000017
3. Wuyep L.C., **Sadaqat H.**, Abdullahi S.L., Micheal E., Ezekiel W.P., Bello R.A., Okalla D., Salman I.I., Bashir M.D., Shallom S.I. (2024). *Encroachment assessment along Kaduna River in Kaduna Metropolis, Nigeria.* Discovery Nature, 1(2), e6dn1027. doi.org/10.54905/disssi.v1i2.e6dn1027

Six further manuscripts in preparation, including the Hunza glacier study (lead author).

LANGUAGES

Urdu (native) · Hindko (fluent) · English (professional working proficiency)